

5.2 What materials should be selected when designing and manufacturing products and prototypes in fashion and textiles?

- a. Understand that most products consist of multiple materials and that fashion and textiles designers are required to discriminate between them appropriately for their use, including:
- natural and synthetic textiles
 - polymers used in component parts, blended textiles
 - metals used for jewellery, component parts and conductive threads
 - wood used for component parts
 - rubber used for performance and functionality.



- b. Demonstrate an understanding of the classification and source of textile fibres and materials, including:
- natural animal textiles, such as:
 - wool, silk and cashmere
 - natural plant textiles, such as:
 - cotton and flax
 - natural mineral textiles, such as:
 - glass fibre
 - synthetic textiles, such as:
 - nylon, polyester and acrylic.



- c. Demonstrate an understanding of the classification of different yarns, including:
- single fibre spun yarns
 - mixed and blended fibre spun yarns, such as cotton/polyester and wool/acrylic
 - filament yarns
 - fancy yarns, such as boucle, chenille and lurex
 - bulked and textured yarns.



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| <p>d. Demonstrate an understanding of the classification of different structures of fabrics, including:</p> <ul style="list-style-type: none"> i. knitted fabrics, including weft, warp and pile knits through hand and machine knitting ii. structured fabrics, such as knotted and braided fabrics/structures, 3D novel structures iii. woven fabrics, such as brocades, jacquards, plaid, tartans and crêpe iv. non-woven fabrics, such as felt, and heated, mechanical and adhesive bonded fabrics v. microfibres. | |
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